

Clinical and Electroencephalographic Follow-up After a First Unprovoked Seizure

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We studied the role of clinical and electroencephalographic factors in the follow-up of children and adolescents after a first unprovoked seizure, and their correlation with recurrence and risk for epilepsy. We conducted a 24-month follow-up of 109 patients aged 1 month to 16 years who had a first unprovoked seizure. We analyzed the characteristics of the first seizure, perinatal history, family history of seizures, electroencephalographic patterns and their influence on seizure recurrence, and calculated risk for subsequent epilepsy. Fifty-six patients (51.4%) had recurrent seizures. The bivariate statistical analysis revealed that maternal prenatal disease (relative risk = 2.02, $P = 0.03$) and an abnormal electroencephalogram (relative risk = 2.89, $P = 0.0003$) were significantly associated with seizure recurrence. Other factors (male sex, partial first seizure, vaginal delivery, family history of seizures, and sleep state) approached statistical significance. Logistic regression revealed that the only variable significantly associated with recurrence was an abnormal electroencephalographic pattern on the first examination (relative risk = 2.48, $P = 0.003$). Cumulative risk ranged from 50-68% at 24 months when the first electroencephalogram was abnormal, and from 26-36% when it was normal. We concluded that the electroencephalogram may have an important diagnostic value in the prognosis of epileptic seizure recurrence in children and adolescents. © 2004 by Elsevier Inc. All rights reserved.

Winckler MIB, Rotta NT. Clinical and electroencephalographic follow-up after a first unprovoked seizure. *Pediatr Neurol* 2004;30:201-206.

Introduction

Seizure and epilepsy are not synonymous: epilepsy refers to recurrent unprovoked seizures [1,2]. The onset of epilepsy usually occurs in childhood, and that explains the concern about the risk for recurrence after the first seizure [3-5]. Several factors predictive of recurrence have been pointed out—age of onset, sex, perinatal and family history, characteristics of the seizure, whether the child was awake or asleep at the time of seizure, and electroencephalographic epileptiform patterns—but no consensus has been reached [6-12].

Few studies about the risk factors for epilepsy [8,10,13] or the incidence and prevalence of epilepsy in Brazil have been conducted so far: some studies revealed prevalence rates of 1/1000 in the general population and 8/1000 among children 5 to 14 years of age [14,15]. Therefore the purpose of this study was to evaluate clinical and electroencephalographic factors associated with the recurrence of seizures in childhood.

Methods

We followed a cohort of 109 children and adolescents age 1 month to 16 years who were consecutively observed 24 to 72 hours after the seizure at the Epilepsy Outpatient Service, Hospital de Clínicas de Porto Alegre, Brazil, from July 1999 to June 2002. Patients were included in the study if they had a first unprovoked seizure or multiple seizures occurring within 24 hours, and if they were not taking antiepileptic drugs. Either one parent or the person responsible for the child signed an informed consent form.

We obtained the full clinical evaluation of patients: demographic data—age, sex, ethnicity, origin; sociocultural data; parents' occupations, parents' and patient's educational level; neurodevelopmental data (according to the assessment of markers of sensory, motor, and cognitive development) [16]; perinatal events—mother's gestational history, type of delivery, Apgar score, postnatal disease; family history of seizures; and characteristics of the seizure—type, duration, patient awake/asleep. Patients underwent physical and neurologic examination, and both awake and sleep electroencephalograms were requested. Patients were invited to join the study group. They were monitored by means of regular visits at

3, 6, 12, 18, and 24 months, in which we investigated recurrence of seizures, clinical events, and introduction and control of antiepileptic drugs.

Awake and sleep electroencephalograms were performed at the first week after the seizure, according to international specifications, and we used either an analog or a digital device with 8, 16, or 24 channels (time constant adjusted to 0.3, tracing speed to 30 mm/s, frequency filter to 70 Hz, and sensitivity between 7 and 10 μ V/mm). Monopolar and bipolar electroencephalographic recordings were obtained with the 10-20 international system of electrode placement.

Electroencephalograms were analyzed by a neurophysiologist from the Hospital who was blinded to the study design. Tracings were classified as normal; with focal or generalized paroxysmal abnormalities, or focal with secondary generalizations; and with unspecific (nonepileptiform) baseline abnormalities [17].

The study was reviewed and approved by the Research and Ethics Scientific Committee at Hospital de Clínicas de Porto Alegre.

Patient referral, contact with families, and clinical evaluation were performed by the authors and resident physicians, with the help of undergraduate medical students from Universidade Federal do Rio Grande do Sul whenever necessary.

Statistical analysis was performed with the software SPSS 8.0 for descriptive analyses—mean, median, mode, standard deviation. We used Cox regression for bivariate analysis to check associations and relative risk of variables and clinical outcome (recurrence) during follow-up. Multivariate Cox regression was performed for factors that had statistical significance values up to 20%. Statistical significance was established at 5% ($P < 0.05$).

Results

Demographic Characteristics

Patients' ages ranged from 1 month to 16 years (mean = 6 years; S.D. = 1 year and 6 months). Most patients were 6 to 12 years of age. Seventy patients (64.2%) were male, and 39 (35.8%) female. Ninety-two patients (84.4%) were Caucasian, and 17 (16.5%), non-Caucasian. The educational level of patients matched age distribution, and the predominant educational level for parents was 8 years (68 parents, 62.4%). Table 1 summarizes the demographic characteristics and relative risk for subsequent epilepsy.

Neurodevelopmental Factors

Ninety-six patients (88.1%) presented normal neurologic development.

Perinatal Events

Reports of events in the prenatal period were made in 26 cases (23.9%): risk of miscarriage in 12 (11%), risk of premature delivery in 5 (4.6%), maternal infection in 2 (1.8%), maternal hypertension in 4 (3.7%), and use of medication in 3 (2.8%). Type of delivery was equally distributed: vaginal in 55 (50.5%) and cesarean in 54 (49.5%). Most patients had a 5-minute Apgar score between 8 and 10 (101, 92.7%), and had adequate weight for gestation (90, 82.6%). In the postnatal period, events reported for 25 (22.9%) patients were as follows: prema-

Table 1. Demographic characteristics, history, neurodevelopment, seizure characteristics, first EEG, and risk for recurrence

	RR	95% CI	P Value
Age (1-24 months)	1.19	0.46-3.04	0.71
Sex (male)	1.51	0.89-2.56	0.12*
Ethnicity (Caucasian)	0.74	0.37-1.48	0.40
Patients under 6 yr of age	1.73	0.98-3.07	0.05 [†]
Parents (<8 yr of education)	1.53	0.51-4.55	0.44
Delayed neurodevelopment	0.97	0.44-2.50	0.95
Prenatal events	2.02	1.11-3.67	0.02 [†]
Vaginal delivery	1.41	0.83-2.40	0.19*
5-minute Apgar score <7	1.41	0.56-3.54	0.46
Low/high weight/GA	1.09	0.56-2.11	0.79
Postnatal events	0.83	0.43-1.62	0.60
Family history of seizures	1.73	0.97-3.10	0.06*
Partial seizures	1.62	0.88-2.98	0.11*
Generalized seizures	0.91	0.05-10.15	0.97
Length up to 5 min	0.57	0.17-1.8	0.36
Length between 5 and 20 min	0.67	0.19-2.40	0.56
Awake state	0.67	0.38-1.19	0.18*
Abnormal EEG	2.89	1.61-5.18	0.0003 [†]
EEG focal paroxysm	2.90	1.71-4.92	0.0001 [†]
EEG generalized paroxysm	1.55	0.73-3.29	0.24

* Statistical significance tendency.

[†] Statistically significant.

Abbreviations:

CI = Confidence interval

EEG = Electroencephalogram

GA = Gestational age

RR = Relative risk

turity, 6 (5.5%); jaundice, 8 (7.3%); infection, 7 (6.4%); disturbance of electrolytes, 4 (3.7%).

Family History of Seizures

Sixty-nine (63.3%) of the 109 patients had a family history of seizures. Table 1 summarizes the neurodevelopmental factors, perinatal events, and family history of seizures, as well as their relative risk for recurrence of seizures.

Characteristics of the First Seizure (Based on Family Report)

Eighty-nine (81.7%) seizures were generalized, and 20 (18.3%) were partial. Generalized seizures were classified as tonic-clonic (46, 51.68%), atonic (20, 22.47%), tonic (19, 21.34%), atypical absence (2, 2.4%), clonic and myoclonic (1, 1.12% each). Partial seizures were homogeneously distributed: simple (6, 30%), complex (7, 35%), and with secondary generalization (7, 35%). Most seizures were short and lasted up to 5 minutes in 83 (76.1%) of the cases. Most children (83, 76.1%) had the first seizure while awake. No one took antiepileptic drugs after the first seizure. The relative risk of these factors for subsequent seizures is presented in Table 1.

Table 2. Epilepsy classification and seizure recurrence

Epilepsy	n	%
Localized	25*	44.64
Idiopathic		
Benign childhood epilepsy with centro-temporal spikes	20	80
Symptomatic		
Temporal lobe epilepsy	2	8
With complex partial seizures, unknown localization	2	8
With partial seizures, secondary generalization, unknown localization	1	1
Generalized	18	32.14
Idiopathic		
Benign myoclonic epilepsy in infancy	1	5.55
Generalized epilepsy with tonic-clonic seizures	17	94.45
Undetermined	13	23.22
Total	56	100.00

* $P = 0.03$

Physical and Neurologic Examination

We observed abnormalities in only 3 (2.8%) patients on neurologic examination, all associated with muscle tone (hypotonia); these abnormalities were not associated with seizure recurrence (relative risk = 1.38; 95% confidence interval, 0.33 to 2.97; $P = 0.65$).

First Electroencephalogram

Findings for the 109 electroencephalograms performed after the first seizure were as follows: normal, 50 (45.9%); with unspecific abnormalities (diffuse slowing or absence of elements characteristic of the sleep phase), 9 (8.3%); with localized paroxysmal abnormalities, 38 (34.9%); with generalized paroxysms, 5 (4.6%); and with localized and secondary generalized paroxysms, 7 (6.4%). Statistical analysis aimed at checking the association of these findings with seizure recurrence is presented in Table 1.

Seizure Recurrence

The rate of seizure recurrence in this study was 51.4% (56 patients) in 24 months of follow-up. Recurrent seizures occurred from 26 hours to 21 months after the first seizure (mean = 4 months and 20 days, S.D. = 5 months; median = 3 months; mode = 2 months). Table 2 presents the epilepsy classification of recurrent seizures according to electroencephalographic results.

Risk for Epilepsy

Multivariate Cox regression was performed for the factors that were statistically significant or approached statistical significance for the recurrence of seizures (Table 3). First electroencephalogram, the main risk factor in our study, revealed influence on seizure recurrence:

when the electroencephalogram was paroxysmal, cumulative risk of recurrence was 50%, 60%, 62%, and 68% at 6, 12, 18, and 24 months, respectively; if normal, risk was 26%, 32%, 34%, and 36% (Fig 1).

Discussion

According to the literature on epilepsy in childhood, the incidence of seizures is highest among Caucasian males 1 to 38 months old [8,10,18-20]. Camfield et al. [21] conducted a population-based study of the incidence of epilepsy in childhood and adolescence in Nova Scotia and found a rate of 48/100,000 in the 1- to 5-year age group, and 43/100,000 in the 6- to 10-year age group, with the lowest rates being observed in infants and adolescents. Hauser [22], however, conducted an epidemiologic study in Chile and reported that the highest incidence of epilepsy was observed among adolescents. Shinnar et al. [23] studied a group of patients 1 month to 19 years of age to investigate risk factors associated with recurrence of

Table 3. Recurrence risk—multivariate Cox regression

Variables	RR	95% CI	P Value
Male	0.76	0.43-1.34	0.35
First partial seizure	1.49	0.80-2.78	0.20
Vaginal delivery	0.74	0.42-1.28	0.28
Prenatal events	1.77	0.95-3.28	0.06*
Family history of seizures	1.29	0.71-2.35	0.39
Sleep state	0.78	0.42-1.45	0.44
First EEG with paroxysm	2.48	1.34-4.59	0.003†

* Statistical significance tendency.

† Statistically significant.

Abbreviations:

CI = Confidence interval

RR = Relative risk

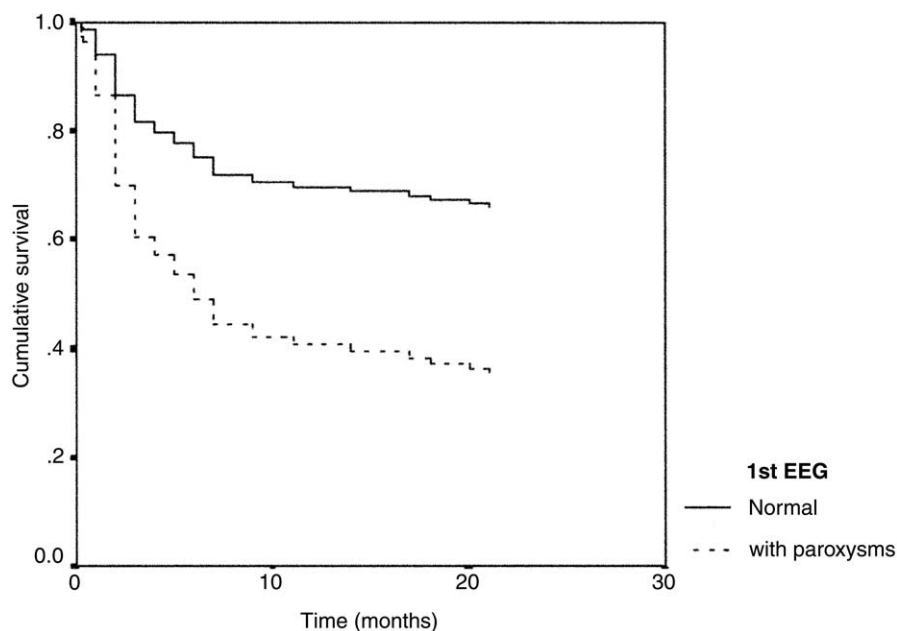


Figure 1. Cumulative risk for seizure recurrence—multivariate Cox regression (relative risk = 2.48; $P = 0.003$).

unprovoked seizures in childhood, and observed that the mean age at first seizure was 6 years and 8 months. In our study, mean age at first seizure was 6 years, and the highest incidence was observed between 7 and 12 years of age, which is in agreement with the findings reported by Camfield et al. [21] and Shinnar et al. [23], but differs from findings by Hauser [22]. Age at first seizure was not associated with recurrence in our study (Table 1).

Reports in the literature agree about the predominance of young males among patients with epilepsy [11,24,25]; our results confirmed that predominance.

The report developed by Instituto Brasileiro de Geografia e Estatística (IBGE, the Brazilian Census Agency) for the year 2001 indicates that there is a predominance of Caucasian individuals in the southern region of Brazil; our findings, thus, reflect the populational distribution in the region, and are not associated with seizure recurrence. Sociocultural characteristics observed in our study also follow the regional populational profile reported by IBGE 2001 [26], and are not associated with epilepsy (Table 1).

Reports about the influence of perinatal events on subsequent epilepsy yield conflicting results [8,11,24,27,28]. Our study revealed that seizure recurrence is significantly associated with a history of maternal prenatal disease, especially risk of miscarriage, the most frequent factor present at our series, which is in agreement with findings reported by Datta and Wirrell [11], although these authors studied a limited age group (Table 1).

Fejerman [29] monitored children with idiopathic partial epilepsy and reported that normal neurodevelopment and neurologic examination indicate a good prognosis. Datta and Wirrell [11], however, reported a relative risk of 2.5 (95% confidence interval, 1.5 to 4.1; $P < 0.02$) for delayed neurodevelopment and seizure recurrence. Most of our patients manifested neither delay in neurodevelop-

ment nor physical/neurologic problems. However, statistical comparison with the studies mentioned above was not possible (Table 1).

Reports in the literature agree about the influence of family history of seizures on seizure recurrence, especially in certain syndromes [8,10,24,27]. We evaluated family history but did not record characteristics of each seizure; family history approached statistical significance for seizure recurrence (Table 1).

Studies of childhood epilepsy conducted in the 1980s reported a predominance of generalized seizures [18,30]. More recently, however, studies with schoolchildren have reported a higher incidence of partial seizures [21,28,31]. We found a predominance of generalized seizures at a 4:1 ratio, in agreement with the findings from the 1980s, perhaps because of the use of similar methodologies. We also found 38 electroencephalograms with focal abnormalities and 25 localized epilepsies; in 13 cases, partial seizures became generalized and were then observed. Winckler and Rotta [8] studied the recurrence of a first provoked seizure and reported a 6 times greater risk for recurrence when the first seizure was partial. Scotoni et al. [10], however, did not find any association between partial seizure and recurrence, results that are in agreement with our results (Table 1).

The short duration of a seizure seems to be an important, although benign, factor for recurrence, that is, it is associated with low risk for subsequent epilepsy [8,32]. We observed a predominance of short seizures, but this variable only approached statistical significance for seizure recurrence (Table 1). The predominance of seizures while the patient is awake has been reported by several authors [8,10,21,23], but the risk for recurrence is greater when seizures occur during sleep [23]. We found a predominance of seizures while awake, but this factor was

not statistically significant for seizure recurrence (Table 1).

The ratio of normal to abnormal electroencephalogram after the first seizure was 1:1 in our study, which agrees with findings reported in the literature [7]. Several authors observed the importance of electroencephalographic epileptiform patterns in the risk for seizure recurrence [7-10,33,34]. Our study confirmed an increased risk for seizure recurrence when the first electroencephalogram was abnormal and, mainly, when there were focal paroxysmal abnormalities (Table 1).

Studies about the recurrence of a first unprovoked seizure in childhood report rates of 33-61% in an average 2-year follow-up [4,8,10,12,23]. The frequency of different forms of epilepsy varies according to the age group studied: for younger children, generalized forms are predominant [8,21,34,35]; for schoolchildren and adolescents, localized forms [24,36-38] are the most common ones. We documented an overall recurrence rate of 51.4% in a 24-month follow-up and no predominant form of epilepsy, perhaps because of the wide age bracket studied (Table 2).

Several aspects should be taken into account when assessing the risk for developing epilepsy to support a decision to start medication. An abnormal electroencephalogram, age at first seizure, physical stress, genetic susceptibility, and a partial first seizure are some of the risk factors reported so far [8,10,32,39,40]. After multivariate analysis, we observed that a paroxysmal electroencephalogram presented a statistically significant risk for seizure recurrence when adjusted to the other variables (Table 3). The cumulative risk for developing epilepsy was reduced in 50% when the first electroencephalogram was normal (Fig 1), which is in agreement with findings reported by Shinnar et al. [23] and Lizana et al. [36].

We found that the incidence of a recurrent unprovoked seizure was 51.4% in 24 months of follow-up; risk factors for the recurrence of seizures, either statistically significant or approaching statistical significance, were history of maternal prenatal disease, family history of seizures, partial first seizure, seizure during sleep, and abnormal first electroencephalogram. The multivariate analysis revealed that only an abnormal electroencephalogram was a statistically significant prognostic factor; cumulative risk was twice as high when electroencephalogram revealed paroxysmal abnormalities.

The authors thank the Conselho Nacional de Desenvolvimento Científico e Tecnológico-CNPQ (Brazilian Agency for Scientific and Technological Development) for the support provided to the development of this study.

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